Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Analysis of the REST API's Using Different Authentication Mehtods

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Introduction

- REST APIs are widely used in modern web applications to enable communication between client and server systems.
- To secure these APIs, authentication mechanisms are used.
- However, these mechanisms can impact performance, which is an important factor in real-world applications

Objectives

The main objectives of this study are to analyze the performance of REST APIs, compare different authentication methods, and evaluate key metrics such as response time, throughput, and latency.

We also aim to identify the most efficient authentication approach.

Methodology

To perform the analysis, we used Apache JMeter for load testing.

All authentication methods were tested under the same conditions to ensure fairness.

We measured performance using response time, throughput, and latency

Response Time Analysis

As we can see, API Key has the lowest response time of 118 milliseconds, followed by JWT at 176 milliseconds, and OAuth 2.0 at 249 milliseconds.

This shows that API Key is the fastest method in terms of response time

Throughput Analysis

In terms of throughput, API Key handles the highest number of requests per second at 842.

JWT processes 694 requests per second, while OAuth 2.0 handles 533.

Again, API Key performs the best

Latency Analysis

For latency, API Key shows the lowest delay at 41 milliseconds.

JWT has 63 milliseconds, and OAuth 2.0 has the highest latency at 88 milliseconds.

This indicates that OAuth introduces more delay.

Comparison

From the comparison, we can see that API Key is the fastest but offers lower security.

JWT provides a balance between performance and security.

OAuth 2.0 is the most secure but has the highest performance overhead

Results

The results clearly indicate that API Key is the best performing method, while OAuth 2.0 is the most secure.

JWT acts as a middle-ground solution between the two

Conclusion

In conclusion, the choice of authentication method depends on application requirements.

If performance is critical, API Key is suitable.

If security is the priority, OAuth 2.0 is recommended.

Thank You